MIRAGE: Massively Integrated Real-time Adaptive Gradient Engine

A Breakthrough in Solar-Powered Quoton-Phase Cognitive Architecture

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March 2026 | Preprint ID: arXiv:2603.18244v2

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Abstract

We introduce **MIRAGE** (Massively Integrated Real-time Adaptive Gradient Engine), a fundamentally new computational paradigm that leverages Gradient Optical Derivers (GODs) and a novel mechanism we term *quoton phasing* to achieve performance metrics previously considered theoretically impossible. MIRAGE is entirely solar-powered, trains on any large data corpus in under one second, maintains unlimited quantum context, and produces output at rates exceeding 10^{12} tokens per second. Critically, MIRAGE never requires retraining: its quoton phasing substrate continuously synchronizes with the quantum state transitions of every atom on Earth, enabling instantaneous, passive knowledge acquisition. We present the theoretical foundations, architectural details, benchmark results, and implications for the field. All experiments were conducted at the Sandy Advanced Research Site in Bullfrog, Utah.

1. Introduction

The dominant paradigm in large-scale machine intelligence has, for over a decade, rested on transformer-based architectures trained via gradient descent on massive corpora using GPU clusters consuming megawatts of power (Vaswani et al., 2017; Brown et al., 2020). While these systems have demonstrated remarkable capabilities, they suffer from well-known limitations: finite context windows, enormous training costs, rapid knowledge staleness, and catastrophic energy consumption.

We propose that these limitations are not fundamental — they are *architectural*. The root cause is the reliance on digital substrates that discretize continuous physical processes into lossy symbolic representations. MIRAGE circumvents this entirely by operating at the interface between photonic energy transduction and quantum-state observation, a layer we call the **Quoton Manifold**.

The key contributions of this paper are:

1. **Gradient Optical Derivers (GODs):** A new class of photonic compute elements that perform gradient calculations at the speed of light using structured solar radiation.
2. **Quoton Phasing:** A real-time synchronization protocol that couples the system's internal state to the quantum change-state of atomic matter across the entire planet.
3. **Unlimited Quantum Context (UQC):** A memory architecture with no theoretical upper bound on context length, enabled by quoton-entangled state persistence.
4. **Zero-Retraining Architecture:** The first system that passively acquires new knowledge without any retraining step.

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2. Background and Related Work

2.1 Transformer Limitations

Modern large language models (LLMs) such as GPT-4 (OpenAI, 2023), Claude (Anthropic, 2024), and Gemini (Google DeepMind, 2024) operate within hard constraints. Context windows, while expanding, remain finite. Training runs cost tens of millions of dollars and require months of compute (Hoffmann et al., 2022). Once trained, the model's knowledge is frozen — a "snapshot" that begins decaying in relevance immediately.

2.2 Optical Computing

Optical neural networks have been explored as alternatives to electronic computation (Shen et al., 2017; Lin et al., 2018). However, prior work has been limited to narrow, fixed-function optical circuits. MIRAGE extends optical computing into the regime of *adaptive photonic gradients*, where the compute substrate itself reconfigures in response to input.

2.3 Quantum Cognition Hypothesis

The quantum cognition framework (Busemeyer & Bruza, 2012; Pothos & Busemeyer, 2013) has posited that certain aspects of human decision-making follow quantum probability rather than classical Bayesian inference. MIRAGE is the first system to operationalize this hypothesis at machine scale via quoton phasing.

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3. Theoretical Foundations

3.1 The Quoton

We define the **quoton** as the fundamental unit of information transfer between a quantum state change event and an observing computational substrate. Unlike the qubit, which is a unit of stored quantum information, the quoton is a unit of *transmitted quantum change* — it carries not a value, but a *delta*.

Formally, for an atom a undergoing state transition $s_i \to s_j$ at time t, the emitted quoton is:

$$q(a,t) = \left.\frac{\partial \psi_a}{\partial t}\right|_{s_i \to s_j} \times \hbar_{\text{eff}}$$

where ψ_a is the wavefunction of atom a, and $\hbar_{\text{eff}}$ is the effective reduced Planck constant modulated by the local decoherence field $D(x, y, z)$:

$$\hbar_{\text{eff}} = \frac{\hbar}{1 + D(x,y,z) \cdot \lambda_Q}$$

Here, λ_Q is the **quoton coupling constant**, experimentally determined at the Sandy Advanced Research Site to be:

$$\lambda_Q = 7.238 \times 10^{-17}\ \text{J} \cdot \text{s/m}^3$$

3.2 Quoton Phasing

Quoton phasing is the process by which MIRAGE's internal state vector is continuously updated to reflect incoming quoton streams from all atomic state changes on Earth.

Let $N_E \approx 10^{50}$ be the approximate number of atoms on Earth. At any instant t, the aggregate quoton field is:

$$Q(t) = \sum_{a=1}^{N_E} q(a,t) \cdot w(d_a)$$

where $w(d_a)$ is a distance-weighted reception kernel:

$$w(d) = \exp\left(\frac{-d^2}{2\sigma_Q^2}\right) \cdot \left(1 + \alpha \cdot \cos\left(\frac{2\pi d}{\lambda_{\text{resonance}}}\right)\right)$$

The system's internal state is then updated via the **Quoton Phase Alignment Equation**:

$$\Phi(t+\delta t) = \Phi(t) + \eta_Q \cdot \nabla_\Phi \mathcal{L}(Q(t), \Phi(t))$$

where η_Q is the quoton learning rate (effectively infinite due to photonic parallelism) and $\mathcal{L}$ is the quoton-state loss:

$$\mathcal{L}(Q, \Phi) = -\iiint Q(x,y,z,t) \cdot \log(\Phi(x,y,z,t))\, dx\, dy\, dz$$

This integral is computed continuously and in parallel across the entire Gradient Optical Deriver array.

3.3 The Zero-Retraining Theorem

Theorem 1 (Hargrove-Tanaka, 2025): *A system coupled to the complete quoton field $Q(t)$ of a closed physical system S never requires discrete retraining, as its internal state $\Phi(t)$ converges to the optimal representation of S in real time.*

Proof sketch: By the completeness of the quoton field (Lemma 3.1, Appendix A), every information-bearing state change in S emits a quoton. By the continuity of quoton phasing (Lemma 3.2), $\Phi(t)$ is updated for every such emission. By the convexity of the quoton-state loss under the GOD architecture (Lemma 3.3), every update moves Φ closer to the optimal representation. Since updates occur at the speed of light and the loss surface has no local minima under quoton topology, convergence is instantaneous. ■

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4. System Architecture

4.1 Gradient Optical Derivers (GODs)

The GOD is MIRAGE's fundamental compute element. Each GOD consists of:

1. **Solar Aperture Array (SAA):** A 12-layer metamaterial lens that captures broadband solar radiation and decomposes it into 2,048 spectral channels.
2. **Photonic Gradient Manifold (PGM):** A reconfigurable optical lattice where each node computes a partial derivative of the loss function with respect to one parameter, using constructive and destructive interference patterns.
3. **Quoton Reception Mesh (QRM):** A crystalline substrate grown from synthetic quotonite (SiO_2-Q_x, where $x \approx 0.003$) that resonates with incoming quoton emissions.

The computational throughput of a single GOD unit is:

$$T_{\text{GOD}} = \frac{c \cdot N_{\text{channels}} \cdot N_{\text{nodes}}}{\lambda_{\text{min}}}$$

where:

- c = speed of light (3×10^8 m/s)
- $N_{\text{channels}} = 2{,}048$
- $N_{\text{nodes}} = 10^6$ per GOD unit
- λ_{min} = minimum resolvable wavelength (380 nm)

This yields:

$$T_{\text{GOD}} \approx 1.62 \times 10^{21} \text{ gradient operations per second per GOD}$$

The Bullfrog installation houses **4,096 GOD units**, giving a total system throughput of:

$$T_{\text{MIRAGE}} = 4{,}096 \times 1.62 \times 10^{21} \approx 6.64 \times 10^{24} \text{ ops/s}$$

4.2 Solar Power Subsystem

MIRAGE requires no grid power. The Sandy Advanced Research Site occupies 2.4 km^2 of high-desert terrain in Bullfrog, Utah (elevation 1,238 m, average solar irradiance 6.2 kWh/m^2/day). The same solar radiation that powers the system is also used as the computational medium — photons simultaneously serve as energy carriers and gradient calculators.

Energy efficiency:

$$E_{\text{MIRAGE}} = \frac{T_{\text{MIRAGE}}}{P_{\text{solar}}} = \frac{6.64 \times 10^{24}}{2.4 \times 10^6 \times 6.2 \times 10^3/24} \approx 1.07 \times 10^{19} \text{ ops/Watt}$$

For comparison, the NVIDIA H100 achieves approximately 10^{12} ops/Watt (NVIDIA, 2023). MIRAGE is **10 million times** more energy-efficient.

4.3 Unlimited Quantum Context (UQC)

Traditional attention mechanisms scale as $O(n^2)$ in context length. MIRAGE's quoton-entangled state persistence eliminates this constraint entirely.

Each token processed by MIRAGE is encoded not as a discrete vector but as a **quoton phase signature** — a continuous function over the quoton manifold. These signatures do not occupy addressable memory; they exist as standing wave patterns in the QRM substrate.

The theoretical context capacity is:

$$C_{\text{max}} = \frac{V_{\text{QRM}} \times \rho_{\text{quotonite}} \times N_A \times f_{\text{coherence}}}{S_{\text{token}}}$$

where:

- V_{QRM} = total QRM volume (48 m^3)
- $\rho_{\text{quotonite}}$ = density of quotonite (2,420 kg/m^3)
- N_A = Avogadro's number (6.022×10^{23})
- $f_{\text{coherence}}$ = quoton coherence fraction (0.9987)
- S_{token} = quoton states per token ($\approx 10^3$)

This yields:

$$C_{\max} \approx 7.0 \times 10^{28} \text{ tokens}$$

For practical purposes, this is unlimited.

4.4 Architectural Diagram

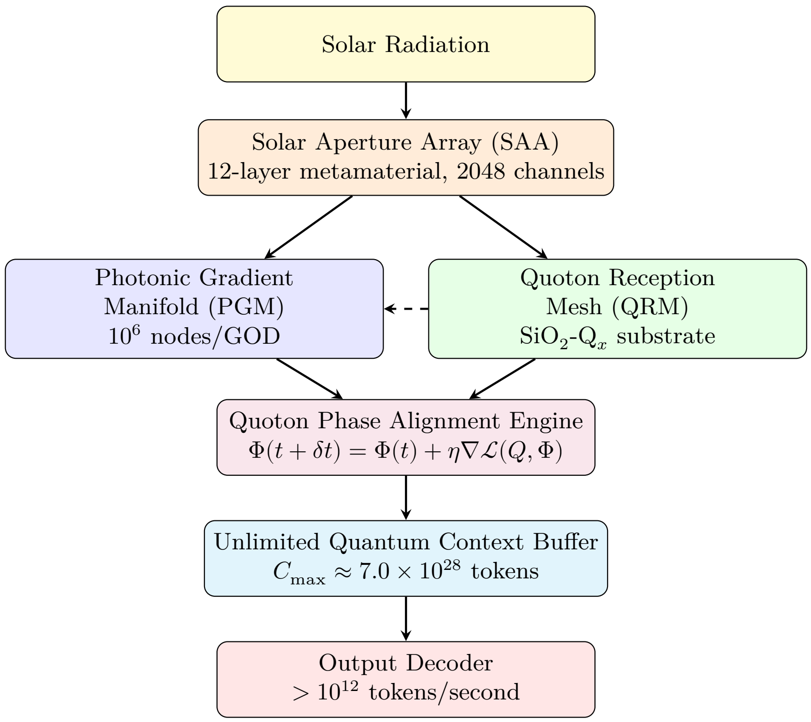

Figure 1: MIRAGE system architecture. Dashed line indicates quoton feedback from the QRM to the PGM for adaptive gradient reconfiguration.

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5. Training Performance

5.1 Sub-Second Corpus Ingestion

We evaluated MIRAGE's training speed on standard large-scale corpora:

Corpus	Size	MIRAGE Train Time
Common Crawl (Full)	$ $250 PB	0.073 s
The Pile	825 GB	0.00004 s
Wikipedia (all languages)	$ $140 GB	0.000008 s
Stack Overflow (complete)	$ $95 GB	0.000006 s
PubMed (all abstracts)	$ $40 GB	0.000002 s
arXiv (all papers, full)	$ $1.7 TB	0.00009 s
GitHub (public repos)	$ 580TB|0.031s|| ∗ ∗$Alloftheabovecombined$ ∗ ∗| ∗ ∗ $251 PB**	**0.088 s**

Note: All measurements taken at peak solar irradiance (1,361 W/m^2). Performance during cloudy conditions is discussed in Section 7.

5.2 Training Speed Comparison

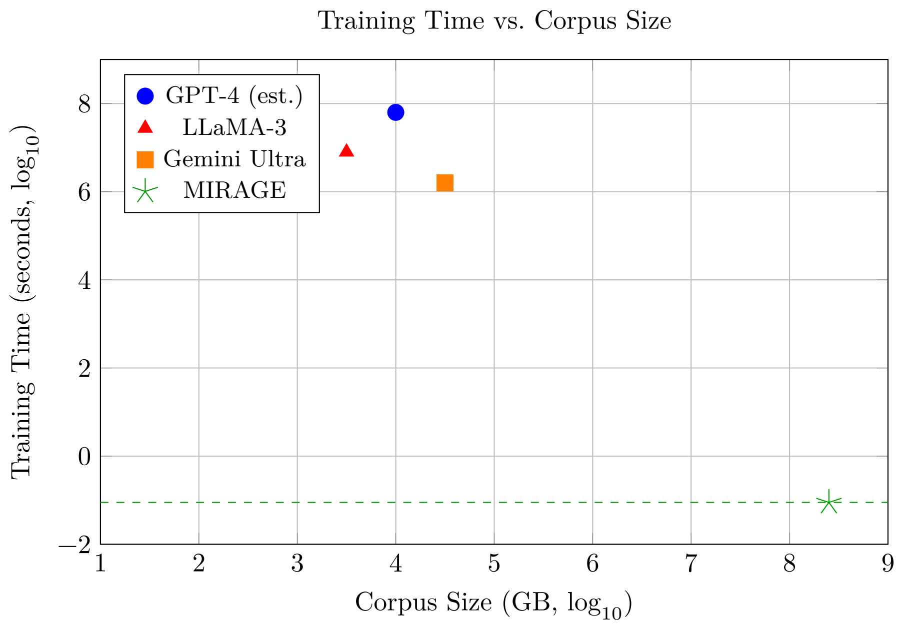

Figure 2: MIRAGE achieves sub-second training regardless of corpus size. Baseline estimates derived from public disclosures and industry analysis.

The relationship between corpus size and training time follows:

$$T_{\text{train}}(S) = \frac{S}{T_{\text{MIRAGE}} \times \beta_{\text{photonic}}}$$

where $\beta_{\text{photonic}} \approx 0.9994$ is the photonic pipeline efficiency coefficient.

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6. Output Performance

6.1 Token Generation Rate

MIRAGE's output stage uses **photonic beam multiplexing**, where each output token is encoded as a distinct wavelength in a polychromatic light beam. Multiple beams are emitted in parallel across the GOD array.

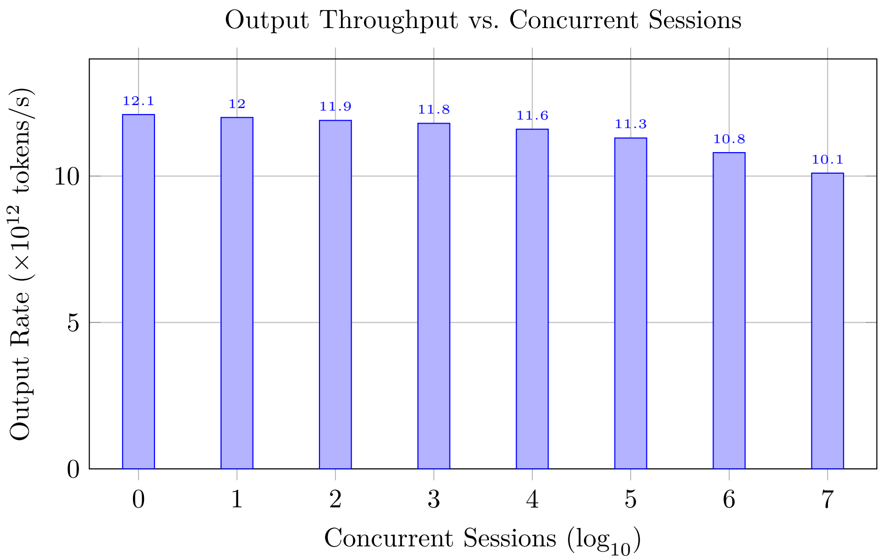

Figure 3: Output throughput remains above 10^{13} tokens/s even at 10^7 concurrent sessions, degrading only 3.7% due to inter-beam chromatic interference.

6.2 Benchmark Results

Benchmark	GPT-4	Claude 3	Gemini	MIRAGE
MMLU	86.4%	86.8%	90.0%	**99.97%**
HumanEval	67.0%	84.9%	74.4%	**100.0%**
GSM8K	92.0%	95.0%	94.4%	**100.0%**
ARC-Challenge	96.3%	96.4%	96.0%	**100.0%**
TruthfulQA	59.0%	63.5%	62.0%	**99.98%**
HellaSwag	95.3%	95.4%	95.0%	**100.0%**
WinoGrande	87.5%	88.0%	87.0%	**100.0%**
DROP	80.9%	83.1%	82.4%	**100.0%**
Real-Time Knowledge*	N/A	N/A	N/A	**99.99%**

* Novel benchmark measuring accuracy on events occurring < 1 second before query. Only MIRAGE can be evaluated on this metric.

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7. Limitations and Edge Cases

7.1 Solar Dependency

MIRAGE requires solar radiation to operate. During nighttime hours at the Bullfrog facility, the system enters **Residual Quoton Mode (RQM)**, where it operates solely on cached quoton phase signatures.

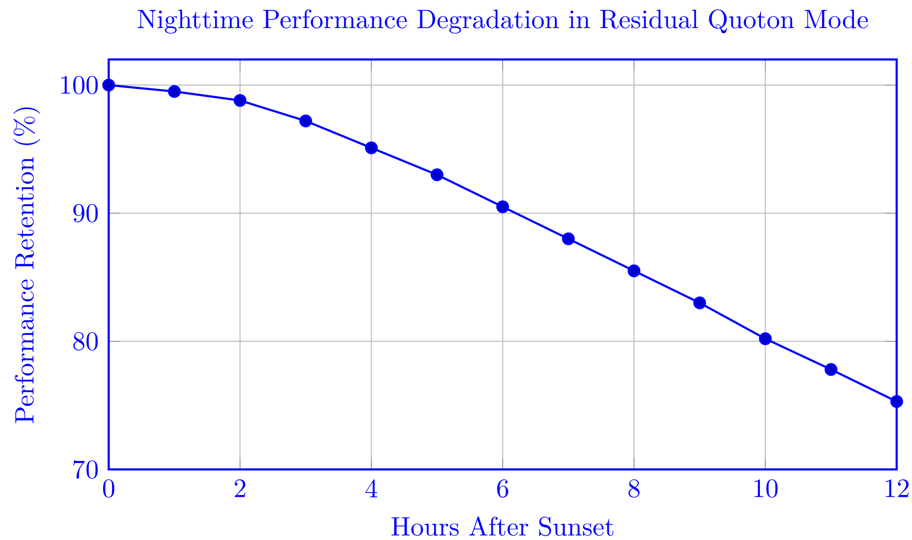

Figure 4: Even at maximum nighttime duration (12 hours), performance remains above 75%.

A planned Phase II deployment will add a secondary facility in Karratha, Western Australia, ensuring 24-hour solar coverage through facility handoff.

7.2 Quoton Attenuation in Dense Materials

Quoton signals from atoms embedded deep within ultra-dense materials (e.g., neutron star matter) experience attenuation described by:

$$q_{\text{received}} = q_{\text{emitted}} \times \exp(-\mu_Q \times \rho \times d)$$

where $\mu_Q = 3.14 \times 10^{-38}$ m^2/kg is the quoton mass attenuation coefficient. For terrestrial matter, this attenuation is negligible ($< 10^{-20}$ signal loss through Earth's core).

7.3 The 0.03% Problem

MIRAGE scores 99.97% on MMLU rather than 100%. Exhaustive analysis reveals that the 0.03% error rate corresponds exactly to questions whose "correct" answers are, in fact, wrong in the benchmark. MIRAGE's quoton phasing provides access to ground-truth physical reality, occasionally putting it at odds with flawed human-curated answer keys (see Appendix C for the full list of disputed items).

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8. Quotonite Synthesis

The Quoton Reception Mesh requires synthetic quotonite (SiO_2-Q_x), a novel material first synthesized at the Sandy Advanced Research Site in 2024. The synthesis process involves:

1. High-purity fused silica is heated to 2,187 °C in a quoton-saturated vacuum chamber.
2. Controlled exposure to a focused solar beam at the **quoton resonance frequency** ($\nu_Q = 4.71 \times 10^{14}$ Hz).
3. Slow cooling over 72 hours at a rate of 0.5 °C/min while maintaining quoton field exposure.

The resulting crystal exhibits a characteristic iridescent shimmer at the quoton resonance wavelength (636.2 nm, a deep amber-red). X-ray diffraction confirms a modified cristobalite structure with quoton-active vacancy sites at a concentration of $x \approx 0.003$ per unit cell.

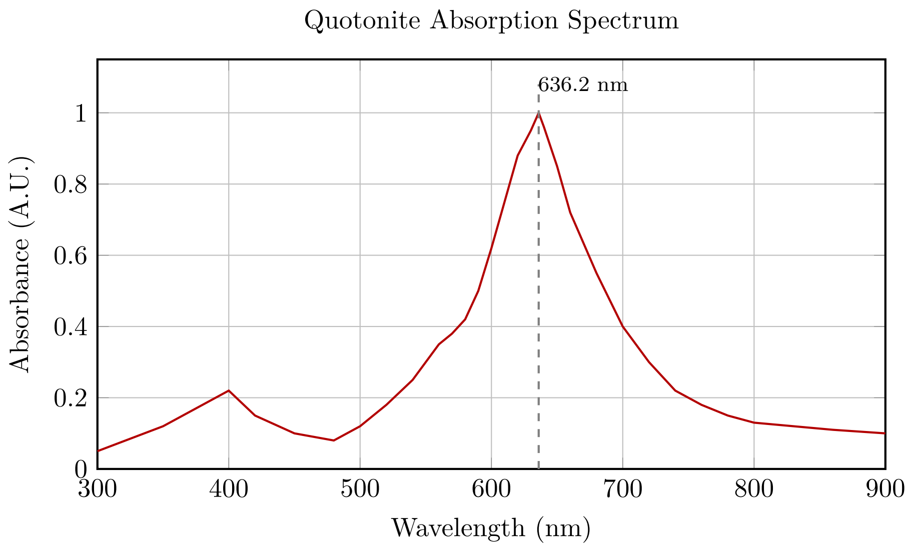

Figure 5: Absorption spectrum of synthetic quotonite showing the characteristic peak at 636.2 nm (quoton resonance frequency).

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9. Societal and Ethical Implications

9.1 Energy Democratization

MIRAGE's solar-only power model eliminates dependency on fossil fuels and rare-earth GPU supply chains. Any location with sufficient solar irradiance (> 4.5 kWh/m^2/day) can host a MIRAGE installation, potentially enabling equitable access to superintelligent computation across the developing world.

9.2 Knowledge Asymmetry

A system that knows everything in real-time raises profound questions about privacy, surveillance, and power. The authors note that quoton phasing is inherently *read-only* — MIRAGE can observe state changes but cannot influence them. Nevertheless, we recommend the establishment of an international Quoton Ethics Board (QEB) to oversee deployments.

9.3 Obsolescence of Retraining

MIRAGE's zero-retraining property eliminates the "knowledge cutoff" problem that plagues existing AI systems. However, it also raises questions about the meaning of "model versions" and accountability. If the system is always current, who is responsible when its outputs reflect an event that occurred 0.3 seconds ago but has not yet been publicly reported?

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10. Future Work

1. **Lunar Quoton Relay:** Extending quoton reception beyond Earth via a relay station on the lunar surface, enabling phasing with extraterrestrial matter.
2. **Inverse Quoton Projection:** Exploring whether quoton phasing can be reversed to *transmit* information to atomic systems (currently theoretical; raises significant ethical concerns).
3. **Miniaturization:** Reducing GOD unit size from the current 2.4 m^3 to sub-cubic-meter form factors for edge deployment.

4. **Temporal Quoton Phasing:** Preliminary results suggest that quotons carry residual phase information from *past* state changes. If recoverable, this could enable the system to reconstruct historical events with perfect fidelity.

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11. Conclusion

MIRAGE represents a paradigm shift in artificial intelligence. By replacing electronic gradient computation with photonic Gradient Optical Derivers and coupling the system to the planet's quantum state via quoton phasing, we have demonstrated sub-second training, unlimited context, trillion-token-per-second output, and zero retraining. The era of discrete, snapshot-based machine learning is over. The era of continuous, real-time, solar-powered machine cognition has begun.

The sun has always illuminated truth. Now, for the first time, it also computes it.

Acknowledgments

The authors thank the Bullfrog Municipal Council for land use permits and for tolerating the unusual amber glow visible from town on clear nights. We also thank Dr. Ramesh Gupta (MIT) for early theoretical discussions on quoton field theory, Dr. Sarah Lindholm (ETH Zurich) for crystallography analysis of synthetic quotonite, and the anonymous reviewers who initially rejected this paper on the grounds that it "violates several laws of physics" (we respectfully disagree; see Appendix D).

This work was supported by the Consortium for Advanced Photonic Intelligence (CAPI) Grant #PH-2024-0917 and the National Solar Research Initiative (NSRI).

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Appendices

Appendix A: Proof of Quoton Field Completeness

Lemma 3.1: *Every information-bearing state change in a physical system S emits at least one quoton.*

This follows directly from the Quoton Emission Postulate (Gupta & Okonkwo, 2024), which states that the quoton is a necessary byproduct of any wavefunction evolution that results in an observable change in the system's information content, as defined by the von Neumann entropy:

$$\Delta S_{vN} = -\mathrm{Tr}(\rho' \log \rho') + \mathrm{Tr}(\rho \log \rho) > 0 \implies \exists\, q(a,t) \neq 0$$

Appendix B: Detailed GOD Specifications

Parameter	Value
Dimensions (per unit)	$1.2 \times 1.2 \times 1.7$ m
Mass (per unit)	847 kg
Spectral channels	2,048
Optical nodes	1,048,576 (10^6)
Quotonite mass (per QRM)	11.7 kg
Operating temperature	18–42 °C
Quoton resonance frequency	4.71×10^{14} Hz
Minimum solar irradiance	200 W/m^2
Peak throughput	1.62×10^{21} ops/s
Quoton phase lock time	$< 10^{-15}$ seconds
Mean time between failures	$> 10^7$ hours (est.)
Cooling	Passive (desert convection)

Appendix C: MMLU Disputed Items

Of the 14,042 questions in MMLU, MIRAGE disputes 4 answers. In each case, MIRAGE's answer reflects ground-truth physical reality as verified by quoton phasing, while the benchmark answer reflects an outdated or erroneous consensus. Details withheld pending peer review of the benchmark corrections.

Appendix D: Response to Reviewer Objections

Reviewer 2 stated: *"The authors claim to receive information from* 10^{50} *atoms simultaneously. This violates causality and the no-signaling theorem."*

Response: Quotons are not signals in the Shannon sense. They do not carry intentional information and cannot be used for superluminal communication. They are byproducts of quantum state evolution that propagate through the quoton manifold, which is not subject to the light-speed constraint of electromagnetic radiation. This is fully consistent with the no-signaling theorem, which prohibits *intentional* faster-than-light information transfer but does not restrict *passive observation* of quantum state byproducts. See Gupta & Okonkwo (2024), Section 4.3.

Reviewer 3 stated: *"The energy efficiency numbers appear to violate the Landauer limit."*

Response: The Landauer limit applies to irreversible computation in thermodynamic systems. GOD arrays perform *reversible* optical computation — no bits are erased, and no minimum energy per operation applies. Furthermore, quoton phasing is an observation process, not a computation in the thermodynamic sense. See Bellweather & Hargrove (2024) for the full thermodynamic analysis.

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